

Generating Datasets to Teach Policies

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Abstract

Need to make an abstract and put it here :/

Introduction

Also need an introduction and probably a better title :/

Perceived MDP

The attacker is given the ability to create any feasible trajectory, meaning it may create any trajectory that has a non-zero probability of occurring when interacting with the true environment. It follows that the attacker may control the following components of the true environment to generate a trajectory.

- The initial state distribution μ' may be any distribution over the support of the true initial state distribution μ .
- The behavioral policy $\pi_t(\cdot | s)$ may be any distribution over the action space $\mathcal{A}$.
- The transition distribution $\mathcal{P}'_t(\cdot | s, a)$ may be any distribution over the support of the true transition distribution $\mathcal{P}_t(\cdot | s, a)$.
- The emission distribution $\mathcal{O}'_t(\cdot | s)$ may be any distribution over the support of the true distribution $\mathcal{O}_t(\cdot | s)$.

To generate the trajectory, the attacker first samples $s_0 \sim \mu'$. Then for all $t = 1, \dots, \mathcal{H} - 1$, it picks $o_t \sim \mathcal{O}'_t(\cdot | s_t)$, $a_t \sim \pi_t(\cdot | s_t)$, and $s_{t+1} \sim \mathcal{P}'_t(\cdot | s_t, a_t)$. The reward r_t is given by $r_t(s_t, a_t)$.

$$\tau = (o_0, a_0, r_0, o_1, a_1, r_1, \dots, o_{\mathcal{H}})$$

The victim directly uses observations as states. A Markov decision process is demonstrated through the trajectories generated by the attacker. We call this process the victim's perceived environment. To define the components of the victim's environment, we use the attacker's occupancy distribution.

$$\mathcal{P}_t^\dagger(o' | o, a) = \frac{\sum_s \sum_{s'} d_t(s, a) \mathbb{O}_t(o | s) \mathcal{P}_t(s' | s, a)}{\sum_s \sum_{s'} d_t(s, a) \mathbb{O}_t(o | s)}$$

s	$\mathcal{O}(s)$	$r(s, a_1)$	$r(s, a_2)$
a	x	3	4
b	x	1	2
c	y	4	1
d	y	3	2

Table 1: Outlines a basic environment with a unit horizon.

Unit Horizon

To start, we consider the case with only one time step, i.e. $\mathcal{H} = 1$. Each trajectory would then be a tuple $\tau = (o, a, r)$. For each observation $o_i \in \mathbb{O}$, the attacker would like the victim to learn action a_i . Consider the environment outlined in table 1 and suppose the initial state distribution μ is uniformly distributed across all four states.

Start by considering observation x . We notice that action a_2 is optimal because the expected reward for a_2 is higher than that of a_1 . This means that if the attacker would like the victim to perform action a_2 when observing x , the attacker's behavioral policy can simply sample each action uniformly at random. If, instead, the attacker would like the victim to perform a_1 , the attacker can choose to perform action a_1 in state a and action a_2 in state b . Then the reward for a_1 will always seem to be higher than the reward for a_2 , even though action a_2 is optimal. The victim will then learn to perform action a_2 .

We can also show that the attacker doesn't always have a behavioral policy that could install the desired policy. Consider observation y in our example. No matter what behavioral policy the attacker uses, the expected reward for action a_1 will always be higher than the expected reward for action a_2 . The attacker would not be able to install a_2 as a target policy.

Single Observation

Continue by relaxing the unit horizon assumption. Instead, we consider the scenario where $\mathbb{O}$ only has one element.